Tushar Nayak

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science, Biomedical Engineering, Research Track — May 2026
Courses: Computer Vision, Learning for 3D Vision, Medical Image Analysis, Image-based Computational Modeling & Analysis.

Manipal Institute of Technology — Manipal, India
Bachelor of Technology, Biomedical Engineering, Minor in Data Science — May 2023

RESEARCH EXPERIENCE

Staff Scientist, Surgical Biomechanics Lab, University of Chicago — Aug 2026 – Present
Patient-Specific Vascular Geometry and Multiphysics Modeling for Geometric Deep Learning based Predictive Modeling

- Lead computer vision & deep learning researcher, investigating relationships between integrated baseline Gaussian curvature, longitudnal curvature adaptive monge patch-wise area growth, and CFD-derived pressure & velocity fields.
- Developing geometry and biomechanics data pipelines and preparing multimodal geometric and physics-based features for subsequent neural network training for predictive modeling.

Research Collaborator, Surreality Lab, University of Pittsburgh — Sep 2026 – Present
Enhanced Intraoperative Speech Analysis for Awake Craniotomy

- Contributing to an automated operating-room speech pipeline combining speaker attribution, word-onset localization, overlap detection, and neural source separation.
- Evaluating overlap detection and selective separation, auditing failure modes and whether enhancement changes quantitative measurements of clinically relevant patient speech.

Student Researcher, Computational Engineering & Robotics Lab, Carnegie Mellon University — Aug 2024 – Jul 2026
Real-Time 3D Vascular Deformation Estimation from Monocular Fluoroscopy for Endovascular Robotic Surgery

- Developed FluoroGRID, a multi-modal deep learning framework combining fluoroscopic input feed, with preoperative vascular geometry, sparse control points, and acquisition metadata to predict 3D deformation.
- Designed a quad-encoder architecture with image features, ordered-chain graph features, sampled mesh geometry, and camera metadata with differentiable projection and physics-based geometric regularizers.
- Generated over 370,000 synthetic dataset with paired 3D–2D examples across views and deformation conditions.
- Achieved real-time inference with sub-millimeter reconstruction precision. Presented initial results at ELSR IROS 2026.

Research Assistant, The ∀ Lab & Image Science Lab, Carnegie Mellon University — Feb 2025 – Aug 2025
Longitudinal Glioblastoma Evolution and MRI-Based Forecasting

- Worked on an Attention U-Net with temporal embedding, and Neural ODE models for continuous-time glioblastoma forecasting from longitudinal multimodal MRI.
- Reduced prediction MSE by 6.7% versus a persistence baseline and relative FLAIR-volume error by 98%; presented at the 2025 CMU Biomedical Engineering Forum.

SELECT PROJECTS

Sparse Echocardiography to 3D Cardiac Volume Reconstruction — Aug 2025 – May 2026

- Reconstructed dense 3D left-ventricular occupancy from three 2D echocardiography slices using a neural implicit representation with Fourier features, differentiable projection and learnable SE(3) slice-pose refinement.
- Achieved 0.864 Dice and 0.765 IoU on MITEA-CAP. Also explored Gaussian occupancy fields (preliminary 0.950 IoU).

Deformable 3D Reconstruction of Endoscopic Surgical Scenes — Apr 2026–Present

- Prototype deformable 3D Gaussian Splatting for monocular endoscopy with camera-aware rendering, depth/pose initialization, deformation fields, and semantic feature alignment.

Geometric Multi-View Breast Cancer Detection — Jan–May 2026

- Built CC/MLO mammography alignment and cross-view attention for lesion localization and classification while leading upstream integration of segmentation and object detection models.
- Achieved reduced lesion-centroid discrepancy by 99.1% and ≈ 0.8 classification accuracy and F1.

Vascular Reconstruction — Nov 2025 – Currently Archived

- Building a preliminary research prototype for sparse-view 3D coronary vessel reconstruction from angiography projections.
- Using Gaussian geometry along with differentiable rendering, and PINN-based hemodynamic regularization.

PUBLICATIONS, MANUSCRIPTS & PRESENTATIONS

Journal Publications

Kite, T., **Nayak, T.**, Jaffee, S., et al. (2026). *The Use of Artificial Intelligence and Machine Learning to Predict Tumor Recurrence in High-Grade Gliomas: A Systematic Review. Neurosurgical Review*, 49(1), 489.

Nayak, T., Gokulkrishnan, N., Chadaga, K., Sampathila, N., et al. (2024). *Automated Histopathological Detection and Classification of Lung Cancer with an Image Pre-processing Pipeline and Spatial Attention with Deep Neural Networks. Cogent Engineering*, 11(1), 2357182.

Mayrose, H., Sampathila, N., Bairy, G. M., **Nayak, T.**, et al. (2024). *An Explainable Artificial Intelligence Integrated System for Automatic Detection of Dengue from Images of Blood Smears Using Transfer Learning. IEEE Access.*

Nayak, T., Chadaga, K., Sampathila, N., Mayrose, H., et al. (2023). *Deep Learning Based Detection of Monkeypox Virus Using Skin Lesion Images. Medicine in Novel Technology and Devices*, 100243.

Nayak, T., Chadaga, K., Sampathila, N., Mayrose, H., et al. (2023). *Detection of Monkeypox from Skin Lesion Images Using Deep Learning Networks and Explainable Artificial Intelligence. Applied Mathematics in Science and Engineering*, 31(1), 2225698.

Peer-reviewed Conference Publications

Nayak, T., Sampathila, N., & Gokulkrishnan, N. (2023). *Processing and Detection of Lung and Colon Cancer from Histopathological Images Using Deep Residual Networks. IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT).*

Gokulkrishnan, N., **Nayak, T.**, & Sampathila, N. (2023). *Deep Learning-Based Analysis of Blood Smear Images for Detection of Acute Lymphoblastic Leukemia. IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT).*

Mayrose, H., Sampathila, N., Bairy, G. M., **Nayak, T.**, et al. (2023). *Deep Learning Approach for Detection of Dengue Fever from the Microscopic Images of Blood Smear. Journal of Physics: Conference Series*, 2571(1), 012005.

Presentations

Nayak, T.*, Basdeo, R*., Andrews, E., et al. (2026). *Toward Real-Time 3D Vascular State Estimation from Monocular Fluoroscopy for Endovascular Robotic Neurosurgery.* 2nd International Workshop on the Evolving Landscape of Surgical Robotics (ELSR), *IEEE/RSJ IROS 2026.*

Nayak, T. & Sampathila, N. (2023). *Automated Oral Squamous Cell Carcinoma Detection from Histopathological Images Using Deep Neural Networks. National Symposium on the Recent Trends in Biomedical Engineering 2023.*

Gokulkrishnan, N., **Nayak, T.**, Sampathila, N., et al. (2023). *Binary Detection of Acute Lymphocytic Leukemia Using Blood Smear Images Using Deep Learning Models. National Symposium on the Recent Trends in Biomedical Engineering 2023.*

Under Review & Manuscripts in Preparation

Basdeo, R., Patel, S., **Nayak, T.**, et al. *A Review of Robotics in Achieving Autonomous and Remote Neuroendovascular Intervention.* Under review, *Journal of Intelligent & Robotic Systems.*

Nayak, T., Basdeo, R., Andrews, E., et al. *Real-Time 3D Neuro-Vascular Deformation Estimation from Sparse 2D Angiography.* Manuscript in preparation for submission to *IPCAI 2027 / International Journal of Computer Assisted Radiology and Surgery (IJCARS).*

RESEARCH EXPERTISE & TECHNICAL SKILLS

3D Computer Vision & Geometry: Sparse-view and multi-view 3D reconstruction, 2D–3D registration, camera and projection geometry, SE(3) pose optimization, geometric correspondence, point-cloud processing, surface mesh reconstruction and deformation, volumetric representations, discrete differential geometry, differentiable rendering and projection.

Deep Computer Vision & Image Processing: Image classification, semantic segmentation, object detection, feature extraction and correspondence, image registration, image restoration, multi-view feature fusion; CNNs, ResNets, U-Nets, attention mechanisms, Transformers, cross-attention, encoder–decoder architectures, spatiotemporal modeling.

Neural Representations & Geometric Deep Learning: Neural implicit representations, coordinate-based networks, Fourier positional encodings, neural occupancy fields, 3D Gaussian Splatting, Gaussian occupancy representations, neural deformation fields, graph neural networks, graph convolutions, message passing, PointNet-style encoders, geometry-aware regularization.

Machine Learning & Computational Modeling: Multimodal learning, few-shot and meta-learning, Neural ODEs, temporal embeddings, continuous-time modeling, differentiable optimization, physics-informed learning, anisotropic diffusion, longitudinal shape analysis, curvature-based geometric descriptors, computational biomechanics.

Medical Imaging & Image-Guided Intervention: CT/CTA, MRI, fluoroscopy/angiography, ultrasound/echocardiography, mammography, endoscopy; medical image segmentation and registration, patient-specific anatomical reconstruction, deformable anatomical modeling, image-guided navigation and intervention.

Programming, Libraries & Research Computing: Python, C++, CUDA, MATLAB, R, Julia; PyTorch, PyTorch3D, PyTorch Geometric, MONAI, TorchDiffEq, TensorFlow/Keras, OpenCV, SimpleITK/ITK, VTK, VMTK, nnU-Net; Linux, Git, Docker, Bash, DICOM, NIfTI.

TEACHING EXPERIENCE

Applied Deep Learning — Computer Science, CMU | Spring 2025
Fundamentals of Computational Biomedical Engineering — Biomedical Engineering, CMU | Fall 2025
Computer Vision for Engineers — Mechanical Engineering, CMU | Fall 2025
Machine Learning in Experimental Biomedical Research — Biomedical Engineering, CMU | Spring 2026

HONORS & AWARDS

CMU Biomedical Engineering Department Head's Fellowship (2024)
Best Paper, 2nd International Conference on Artificial Intelligence, Computational Electronics and Communication (2023).